

CO7093 - Big Data & Predictive Analytics

CW Assignment Classification & Clustering

Assessment Information

Contribution to overall mark	70%
Date Released	13/02/2026
Submission Deadline	08/04/2026, 12:00PM

Overview

The primary aim of this assignment is to guide you through the essential stages of a data science project. You will learn to formulate meaningful questions, explore and analyze data, visualize key insights, and develop predictive models to generate new predictions. Throughout this process, you will apply the concepts, techniques, and skills acquired during the module. The dataset for this assignment is provided in CSV format.

Learning Objectives

After this homework, students will be able to:

- Work effectively with fundamental Python data structures such as dictionaries, tuples, and lists.
- Use **Pandas** as the primary tool for processing structured data in CSV files, including handling edge cases and applying appropriate techniques to manage missing data.
- Create exploratory visualizations using **Matplotlib (PyPlot)** and extend these skills toward producing more advanced visualizations using **Seaborn**.
- Train machine learning models with **scikit-learn** and use them to generate predictions for future or unseen data.
- Utilize **PySpark** to explore, analyse, and demonstrate efficient methods for processing and managing large-scale datasets.

How to submit

For this assignment, you need to submit the followings:

1. **A short report** (about 8-10 pages in pdf including any images) on your findings in exploring the given dataset, a description of your model and its evaluation, a description of your clusters and its justification, local classifiers based on your clusters as well as their evaluation.

2. **The Python source code** written to complete the tasks set in the paper. You should submit your solution either as ipynb file (e.g., group1_solution.ipynb) or py file (eg., group1_solution.py). Note that even if you decide to work on your own, you must enrol yourself into a group.
3. **Self-Assessment Form** – This should include an estimate of time and effort invested by individual group members.

Please put your source code, report and signed self-assessment form into a zip file CW2_GroupID.zip (e.g., CW2_Group1.zip) and then submit your assignment through the module's Blackboard page by the deadline. Note that to submit, you need to click on the Coursework link on Blackboard and then upload your zipped file. Remember it is **1 submission per group**. You should use the **GitLab** version control system to manage your code and report. Ensure that you make at least three meaningful commits representing different stages of your work: (1) Exploratory Data Analysis (EDA), (2) Model Development, and (3) Distributed Analysis. For every commit, upload a document, clearly outlining the contributions of each group member. Detailed instructions on using GitLab will be provided shortly.

Academic Misconduct

This is a group assignment, and collaboration with other groups is not permitted.



This project is in the “Amber” category. You may use generative AI to support your understanding and preparation, but you must NOT submit AI-generated code, text, analysis, or visualisations as your own work. Any AI assistance must be advisory only.

You may use AI to:

- Clarify what the project tasks are asking you to do.
- Understand Python error messages or unexpected behaviour in libraries such as Pandas, Matplotlib, scikit-learn, or PySpark.
- Explore alternative ways of approaching a data-analysis task (e.g., different strategies for handling missing data, feature engineering, splitting datasets, etc.), as long as you make the final decisions yourself.
- Seek advice on the advantages and disadvantages of different modelling, visualisation, or data-processing choices.
- Generate ideas for new test cases to validate data-processing or modelling logic.
- Get conceptual explanations of machine-learning algorithms or big-data processing patterns.

You must not (this list is not exhaustive):

- Upload the full project description or dataset to an AI tool and ask it to complete the work for you.
- Upload your code, notebook, or analysis and ask AI to “fix it” or “complete it.”
- Ask AI to write Python, Pandas, scikit-learn, Matplotlib/Seaborn, or PySpark code for you to submit.
- Ask AI to choose your modelling approach, visualisation strategy, feature-engineering steps, or PySpark pipeline for you.
- Use AI “agents” or tools to automatically generate substantial portions of your analysis, visualisations, or predictive modelling workflow.

Rule of thumb:

You may use AI in the same way you would discuss the project with another student not in your group—to talk through ideas, concepts, errors, or possible approaches—but not to obtain code or solutions to submit.

Any breach of this policy constitutes academic misconduct.

If you have used any generative AI in relation to this assignment, you must include in your submission (either as comments in the code or in the report), a brief explanation of exactly which AI you used, and how you used it (1-2 sentences would do). Absence of any such note will be treated as a declaration that you did not use AI. You are not required to submit any further details of AI use such as prompts and chat history, but you should preserve such information, and in the event of an academic misconduct investigation we may demand to see such evidence.

This is a group assignment, but collaboration between groups is not permitted. Plagiarism will be treated strictly according to standard university and departmental procedures. Your submissions will be sent to a plagiarism detection service. In line with university policy, marking will be done anonymously. Only the Blackboard-supplied userid / student number will be visible in marking. For the above two reasons, do not include your name or any other personally identifiable information in your report or code.

Instructions for group work

This assignment may be attempted in groups of size up to 3. Group size of 1 (individual groups) are also allowed. The link to join a group will be available on the module’s Blackboard page under Assessment and Feedback Section. Please read the following instructions carefully before you join a group.

1. The maximum number of members in a single group are 3.
2. To join a group, please complete the form available on Blackboard.
3. The deadline to join a group is 27/02/2026 at 12:00. After the deadline you will be automatically assigned to your own group of size 1.
4. Before joining a group, please discuss it with other group members and make sure that you are happy working with each other.

5. Once the groups are finalised, you will not be able to leave or join another group.

Normally, a group will be given the same mark unless some members made little or no contributions. Any group can be called for an interview during marking. All group members **must attend**, explain their contributions, and defend the work submitted.

Problem Statement

In this coursework, you will create classification models that can predict whether a patient will readmit to the hospital within 30 days using 130 US hospitals diabetes dataset. This dataset represents ten years of clinical care at 130 US hospitals and integrated delivery networks. Each row concerns hospital records of patients diagnosed with diabetes, who underwent laboratory, medications, and stayed up to 14 days. The dataset has 47 features and 101766 instances. The dataset is provided on the blackboard.

Objective

Using the given dataset, the goal is to determine if the patient is at high risk and will be admitted to ICU or not. You will use appropriate performance metrics to evaluate the performance of your model. The data is not clean, and you will have to apply appropriate methods to clean the data. Additionally, using unsupervised clustering, you will have to implement cluster-based classification model that may improve the performance of the model. The (partially) processed dataset is available to download from the blackboard.

PART 1: DATA EXPLORATION AND PREPROCESSING

Your first task is to prepare the data and carry out data cleansing, bearing in mind the question you would like to answer. For example, which factor is the most important factor in predicting the readmission of a patient.

A. Prepare and clean the data:

If you have a closer look at the dataset, you will see that there are lots of inconsistencies and missing values data. You need to deal with those missing values appropriately but in the first instance, you should consider an aggressive approach to deal with them. Following is a list of some of the steps that you may consider.

- Begin by exploring the structure of the dataset. For example, you might inspect its shape, the types of variables, and any initial patterns you observe.
- Consider whether any columns serve no meaningful analytical purpose. For instance, identifier-only features (e.g., encounter_id) may need to be removed. Justify your decision.
- Investigate the presence of missing or irregular values. Some missing entries may be encoded using non-standard placeholders (such as '?'). Identify such cases and decide how to standardise them (e.g., replacing them with NaN).

- Summarise the extent of missingness in the dataset. Reflect on how this summary changes after applying your cleaning steps.
- Examine the target variable (readmitted). Since the objective relates to predicting early readmission, propose and implement an appropriate way to convert this variable into a useful form (e.g., binary encoding).
- Review the datatypes of all columns and assess whether any conversions or adjustments are necessary for analysis or modelling.
- Calculate the proportion of missing values per column and decide which features, if any, should be removed due to excessive missingness. Provide justification for any thresholds you choose.
- Identify features with little or no variability. Determine which columns (such as those containing only a single repeated value) are unlikely to contribute useful predictive information, and consider removing them.
- Explore the dataset for near-zero-variance features. Decide whether any such columns should be excluded from further modelling and explain your reasoning.
- Generate summary statistics for the numerical features. Assess whether there are potential outliers and formulate a defensible strategy for handling them.
- Evaluate whether feature scaling or normalisation is required for your chosen modelling techniques and apply it as appropriate.
- Conclude this stage of your analysis by reviewing how the dataset has changed (e.g., by checking its final shape or summarising the cleaned feature set).

B. Data Visualization

Using the cleaned and transformed DataFrame, explore meaningful relationships and patterns within the dataset. Your visualisations should help you better understand the characteristics of the features and their potential relevance to prediction.

- Begin by examining the distribution of the target variable (readmitted). Create an appropriate plot to illustrate how frequently each class occurs.
- Investigate how readmission rates vary across different age groups. Visualise the relationship in a way that allows you to comment on any noticeable trends.
- Explore the connection between the number of medications and the likelihood of readmission. Create a plot that clearly communicates how the target variable behaves across different medication counts.
- Produce broader exploratory visualisations, such as a scatter-plot matrix or a correlation heatmap. Although these may be large, they can help highlight pairs of features with particularly strong relationships. Identify and comment on any highly correlated feature pairs.
- Extend your analysis by creating additional visualisations of your choice. Use these to demonstrate your understanding of the data and the problem context.
- Use libraries such as **Seaborn** to produce more informative and aesthetically clear visualisations.

PART 2: INITIAL MODEL BUILDING

Feature Selection and Model Development:

Explore the dataset to identify predictors that are most relevant for modelling. Justify your feature-selection choices based on domain understanding, statistical properties, or exploratory analysis. Using the classifiers introduced in the module, develop any three different classification models you consider suitable for the task.

Data Splitting and Validation:

Partition the dataset into appropriate training and testing subsets. Implement a suitable cross-validation strategy (e.g., k-fold cross-validation) to obtain a reliable estimate of each model's performance, and explain why the chosen validation approach is appropriate.

Handling Class Imbalance:

Address the imbalance in the target variable by selecting all examples from the minority class and an equal number of randomly selected instances from the majority class using random undersampling.

Performance Evaluation and Comparison:

Evaluate each model using appropriate metrics (e.g., accuracy, precision, recall, F1-score, ROC-AUC). Compare and interpret the results to identify which model performs best under the chosen conditions. Reflect on the strengths and limitations of each modelling approach.

PART 3: DISTRIBUTED ANALYSIS

Distributed Model Development:

Use PySpark's distributed computing capabilities to build an enhanced classification model using the full dataset. Apply and justify appropriate strategies for handling missing or incomplete data to maintain data quality at scale. Reflect on how distributed processing influences model training and performance.

Clustering Analysis:

Perform clustering on the cleaned dataset using K-Means or another suitable clustering algorithm. Determine an appropriate number of clusters by applying and explaining methods such as the elbow method, silhouette analysis, or other relevant criteria. Provide clear and insightful visualisations to illustrate the structure and separation of the resulting clusters.

Cluster-Based Classification:

Develop local classifiers within each cluster and evaluate their performance. Compare these results with the performance of the global (non-clustered) classification model. Discuss whether the cluster-based approach leads to improved predictive performance and under what conditions such an approach may be beneficial.

Marking Criteria

The following areas are assessed:

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|---|------------|
| 1. Cleansing, visualizing, and understanding the data. | [30 marks] |
| 2. Building up and evaluating the predictive model. | [20 marks] |
| 3. Improved model, Clustering and evaluating cluster-based model. | [20 marks] |
| 4. Coding style and use of appropriate data structures. | [10 marks] |
| 5. Writing the report and interpreting the results. | [20 marks] |

Indicative weights on the assessed learning outcomes are given above and can be found in the marking rubric on Blackboard. The marks are given according to the following guidelines.

- **[90 – 100]:** A predictive model with excellent performance, excellent justification and visualisation of the clusters, great insights from the data, and a report of professional standards, while leveraging PySpark for big data analysis.
- **[80 – 90]:** A comprehensive coverage of data cleansing techniques demonstrating an excellent understanding of the data, a sound comparison of the global predictive model against the clusters-based model, appropriate use of PySpark, and a well-structured, maintainable, and robust code usefully using functions.
- **[70 – 80]:** As in Second Upper plus a well-justified predictive model by the data cleansing with good performance using sound evaluation techniques; a well-justified clusters and a concise report on the results obtained from the dataset. Demonstration of some basic understanding of effective use of PySpark for analysing big data.
- **[60 – 70]:** A good coverage of data cleansing techniques exploring the dataset, a good visualisation of the clusters, a predictive model with an appreciable accuracy with a rationale behind it, a working code and a well- structured report on the results obtained from the dataset.
- **[50 – 60]:** Essential data cleansing techniques are covered, a predictive model partially justified with an appreciable accuracy, a working clustering, a partially commented code with very few functions, and a narrative of the findings about the dataset with few deficiencies.
- **[20 – 50]:** Data is not cleaned appropriately. Some important features are ignored. Dataset is too small, and the model is overfitting the data.
- **[0 – 20]:** Code is not complete and is not compiling. Report is incomplete.